Project Smart Fake News Detection System
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Program: AI/ML with Python

## 1. Problem Statement

There is a lot of information on the internet. This makes it important to have systems that can check if text is true or not. This project creates a system that uses Natural Language Processing and machine learning to tell if text is news or fake news. The system can be used on a website where people can see the results easily.

## 2. Dataset Description

There is a collection of text that includes headlines and articles. Each item, in the collection is marked with a number. The number 1 means it is news. The number 0 means it is fake news.

## 3. Data Preprocessing & Feature Engineering

The text was. Changed into numbers. A tool called TF-IDF Vectorizer was used for this. The words that are not important were removed. This helped make the data better for the machine learning system to use.

## 4. Machine Learning Methodology

The data was given to a model called Logistic Regression. This model is good because it works fast and is easy to understand. The model and the tool that changed the text into numbers were saved so they could be used later.

## 5. Application Deployment

The model was put into a website made with Streamlit. People can use this website by typing in text. The website then shows if the text is real or fake. The website makes it easy for people to use the system.

## 6. Results & Conclusion

The system worked perfectly on the test data. This was because the real and fake news had different ways of writing. The website works well. Helps people use the machine learning system without any problems.